

ConnectTel Customer Churn Prediction

End-to-End Machine Learning Project Report

Dataset: IBM Telco Customer Churn | 7,043 customers | 21 features
Models: Logistic Regression | Random Forest | XGBoost
Best AUC-ROC: 0.8423 | Recall (Churn): 80%

1. Project Overview

ConnectTel is a telecommunications company facing significant customer churn. This project builds a machine learning pipeline to identify customers at high risk of churning, enabling the business to take proactive retention actions.

Dataset : IBM Telco Customer Churn (7,043 customers, 21 features).
Target : Churn (Yes / No).
Churn rate: 26.54% (imbalanced binary classification problem).

Pipeline:

- * Data loading & inspection
- * Exploratory Data Analysis (EDA)
- * Feature engineering & preprocessing
- * Model training (Logistic Regression, Random Forest, XGBoost)
- * Model evaluation (AUC-ROC, Precision, Recall, F1)
- * Explainability (SHAP values)
- * Strategic business recommendations

2. Business Problem

Customer acquisition costs 5-25x more than retention. In telecom, annual churn rates of 15-25% represent a severe revenue drain.

By predicting which customers are likely to churn, ConnectTel can:

- * Target at-risk customers with personalised retention offers
- * Reduce revenue loss from involuntary cancellations
- * Improve overall Customer Lifetime Value (CLV)
- * Allocate retention budgets more efficiently

Key success criterion: AUC-ROC ≥ 0.80 with high recall on the churn class.

3. Key Dataset Statistics

Total customers : 7,043
Features (raw) : 21
Churn rate : 26.54% (Yes=1869, No=5174)
Missing values : 11 rows in TotalCharges (new customers, tenure=0)

Numeric features:

tenure -- mean 32 months, max 72 months

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MonthlyCharges -- mean \$64.76, max \$118.75

TotalCharges -- mean \$2,283, max \$8,684

Engineered features:

TotalChargesPerTenure -- average monthly revenue per customer

ServiceCount -- count of value-add services subscribed

4. EDA Insights

Insight 1 -- Contract Type (Strongest Predictor)

Month-to-Month ~42% churn rate

One-Year ~11% churn rate

Two-Year ~3% churn rate

=> Monthly subscribers are ~14x more likely to churn than two-year subscribers.

Insight 2 -- Internet Service Type

Fiber Optic ~42% churn rate (high value, high dissatisfaction)

DSL ~19% churn rate

No Internet ~7% churn rate

Insight 3 -- Tenure Drop-Off

~50% of churners leave within the first 12 months.

Churn risk declines steeply after 24 months of tenure.

New customers (<1 year) are the highest-risk cohort.

Additional findings:

- * Electronic check payment correlates with higher churn.
- * Senior citizens churn at a slightly higher rate.
- * Customers without partners/dependents churn more often.
- * Lack of OnlineSecurity & TechSupport increases churn risk.

5. Preprocessing & Feature Engineering

Data Cleaning:

- * TotalCharges converted from object to numeric; 11 NaN rows filled with 0.
- * customerID column dropped (non-predictive identifier).
- * Target variable Churn mapped: Yes -> 1, No -> 0.

Feature Engineering:

- * TotalChargesPerTenure = TotalCharges / max(tenure, 1)
- * ServiceCount = count of 'Yes' across 6 add-on service features

Encoding:

- * All categorical variables one-hot encoded (drop_first=True).
- * Final feature matrix: 32 features.

Train/Test Split:

- * 80% training (5,634 samples), 20% testing (1,409 samples).
- * Stratified split preserves 26.54% churn ratio in both sets.

Scaling:

- * StandardScaler applied to 5 numeric features.
- * Scaler fitted on training data only -- no data leakage.

6. Model Comparison

Three models trained with class-balanced weighting (imbalanced target).

Logistic Regression:

AUC-ROC : 0.8417
Recall (Churn) : 78%
Precision (Churn) : 50%
F1 (Churn) : 0.61
Accuracy : 74%

Random Forest (n_estimators=150, max_depth=8, class_weight='balanced'):

AUC-ROC : 0.8423
CV AUC-ROC (5-Fold): 0.8454 +/- 0.0115
Recall (Churn) : 80%
Precision (Churn) : 53%
F1 (Churn) : 0.64
Accuracy : 76%

XGBoost (GridSearchCV best: lr=0.1, max_depth=3, n_estimators=100, subsample=1.0):

AUC-ROC : 0.8423
Recall (Churn) : 80%
Precision (Churn) : 51%
F1 (Churn) : 0.62
Accuracy : 74%

Conclusion:

All models exceed the AUC-ROC 0.80 benchmark.
Random Forest is the recommended model (highest recall & F1 on churn class).
Recall (catching actual churners) is prioritised over precision.

7. Explainability -- SHAP Analysis

SHAP (SHapley Additive exPlanations) was used to interpret the Random Forest model.

Global Feature Importance (Beeswarm Plot):

- #1 Contract_Month-to-month -- strongest positive driver of churn
- #2 tenure -- long tenure strongly reduces risk
- #3 InternetService_Fiber optic -- Fiber subscribers show elevated risk
- #4 TotalChargesPerTenure -- higher cost per month increases risk
- #5 OnlineSecurity_No -- absence of security services raises risk

Individual Prediction (Waterfall Plot):

For a single high-risk customer, SHAP decomposes the prediction into individual feature contributions, making the model auditable and actionable for customer service teams.

Business Value:

SHAP allows ConnectTel to explain predictions to stakeholders, comply with explainability requirements, and tailor interventions.

8. Strategic Recommendations

Recommendation 1 -- Contract Migration Campaign

- * Target Month-to-Month customers with tenure <12 months and predicted churn probability > 0.60.
- * Offer discounted upgrade to 1- or 2-Year contracts.
- * Expected impact: 15-20% reduction in early-tenure churn.

Recommendation 2 -- Fiber Optic Satisfaction Programme

- * Proactively survey Fiber Optic customers monthly.
- * Bundle OnlineSecurity and TechSupport at preferential rates.
- * Offer loyalty pricing after 6 months.
- * Target: reduce Fiber churn rate from ~42% to ~28%.

Recommendation 3 -- First-Year Onboarding & Nurture

- * Structured onboarding for all new customers (months 1-12).
- * Automated retention offers at 3-month & 6-month tenure marks.
- * Escalate NPS detractors immediately.

Recommendation 4 -- Electronic Check Migration

- * Incentivise auto-pay (bank transfer / credit card) -- lower churn.

Recommendation 5 -- Senior Citizen Retention

- * Dedicated support programme for senior citizens (16% of base).
- * Simplified billing and dedicated phone support channel.

9. Conclusion

This project successfully built a churn prediction system for ConnectTel achieving AUC-ROC of 0.8423 with 80% recall on the churn class.

Key findings:

- * Contract type, tenure, and internet service are the primary churn drivers.
- * Feature engineering improved model quality and explainability.
- * SHAP bridges the gap between ML predictions and business actions.
- * Random Forest is recommended as the production model.

Next steps:

- * Deploy the model as a REST API for real-time scoring.
- * Integrate with CRM for automated retention workflow triggers.
- * Retrain monthly as new customer data accumulates.
- * A/B test retention campaigns and measure uplift vs. control group.